

Empowering Civil Servants: Fostering Ethical Reflection on GenAI Use through Experimentation and a Serious Game

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Abstract

While there is much focus on interventions to foster ethical reflection in the design process of AI, there is less focus on fostering ethical reflection for (end)users. Yet, with the rise of genAI, AI technologies are no longer confined to expert users; non-experts are widely using these technologies. In this case study in a governmental organization in the Netherlands, we investigated a bottom-up approach to foster ethical reflection on the use of genAI tools. An approach of guided experimentation, including an intervention with a serious game, allowed civil servants to experiment, to understand the technology and its associated risks. The case study demonstrates that this approach enhances the awareness of possibilities and limitations, and the ethical considerations, of genAI usage. By analyzing usage statistics, we estimated the organization's energy consumption.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

responsible AI, industry practice, genAI use, serious game, public services

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1 Introduction

With the rise of generative AI (genAI), AI technologies are no longer confined to expert users or specialized domains; they are now

being widely adopted by non-experts across various sectors. Non-technical individuals (without programming skills or understanding of AI) are increasingly interacting with these tools in their natural language. This leads to both opportunities – as a larger group can benefit from the accessibility and ease of use of these technologies – and risks, as a lack of understanding how these tools operate may lead to an overestimation of their capabilities [8].

In the public domain, as well as in society at large, striking a balance between the potential value of these new innovative tools and the risks of uncritical implementation has become a significant challenge [6]. Some governments and public organizations have deemed the risks to outweigh the benefits and decided to ban the use of genAI tools altogether (Italy, for instance, banned the use of ChatGPT [19]). Other governmental organizations take a more liberal approach and prefer to inform and encourage civil servants to take ownership of their actions. This approach promotes autonomy and responsibility, allowing civil servants to use genAI tools, while remaining accountable for their actions.

Governmental organizations in Europe have responsibilities due to legal requirements such as the data privacy regulations (GDPR) and, nationally, the civil service oath for handling data and information. Furthermore, as a public organization they need to uphold the principle of transparency when dealing with all decisions relevant for citizens. After the introduction of genAI tools, it became unclear how to fulfill these responsibilities. There were no general and shared guidelines for governmental organizations on how to handle genAI tools. There was a lack of insight into how employees prompt these tools, how they apply the generated output in their work, and, more critically, their awareness of the associated risks. As the viewpoints and behaviors of employees regarding these tools were unclear, it was difficult to assess whether the tools were used responsibly, or to establish a shared understanding of what responsible use entails.

To investigate and explore these questions, we conducted a pilot study within a governmental organization in the Netherlands. In the pilot we investigated a bottom-up approach of guided experimentation to promote critical and self-aware use of genAI tools among civil servants. As one of the interventions we developed a serious game to encourage and guide ethical reflection on the use of genAI.

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2 Moving from theory to practice

Earlier research already outlined the risks of uncritical implementation of genAI tools within society. The risks mentioned most in the literature are data insecurity [8, 17, 22] and the fallibility of LLMs [4, 6, 17]. Beltran et al. [3] systematically analyzed guidelines from national governments worldwide and compared the foreseen risks of genAI adoption in the public sector. The predominant common concern they found was government confidential information leakage. Other risks often mentioned were the potential for bias and hallucination, cybersecurity, and privacy risks. Finally, environmental considerations were mentioned in some guidelines. Remarkably, this issue was lacking in most national governments' guidelines.

While IT/AI- policymakers and advisors may have awareness and concerns about the potential risks of genAI adoption, it is the employees on the workforce who have the most influence on this and interact most with these tools. Through guidelines, policymakers and advisors aim to set norms in an organization. However, it is well known that organizational frameworks and guidelines are not always effective to operationalize certain behaviors [23]. In practice, guidelines are often considered too abstract to be action-guiding, they sometimes give conflicting or ambiguous guidance, and in the worst case, they are simply not read.

This gap between theory and practice is well-known in the field of AI ethics [2, 13, 16]. With the surge of modern technologies, philosophers, sociologists and policymakers all contribute to the discussion on the wider consequences of the implementation of these technologies in our society. However, in many cases, there is a gap between theorists discussing the impact of technology on the one hand, and practitioners – developing or using these technologies – on the other [14]. Ideally, it should not only be academics and policymakers reflecting on the impact of technology, but all stakeholders including end-users should be aware of the potential risks involved, to balance the trade-offs between the risks and the potential benefits.

2.1 Interventions and bottom-up approach

Various researchers and practitioners in academia, industry and civil society organizations investigated how to close the gap between theory and practice. This yielded various overviews, mappings and toolkits of methodologies and interventions available to promote ethical reflection in the implementation of new (AI) technologies [13, 24]. Most existing ethical tools and frameworks focus on developers of AI technologies. For instance, in a systematic analysis of AI ethics tools [2], most interventions were aimed at promoting and including ethical considerations in the design and development process. Out of the 39 tools and frameworks the researchers investigated, only two explicitly mentioned 'users' as a stakeholder group. Wong et al. [24] also stressed that most toolkits implementing AI ethics in practice focus on individual technical practitioners. Yet, there seem to be little tools or interventions available that guide ethical decision-making specifically aimed at end-users and teams working with, rather than working on, AI-applications.

In this pilot study, we aimed to create an interactive approach to promote the critical and responsible use of generative artificial intelligence (genAI) among civil servants. As an intervention, we focused on designing an approach to encourage experimentation and

ethical reflection, leading to a more comprehensive understanding of the technology and its ethical implications. In this, we drew inspiration from researchers and designers, who emphasized interactive and/or playful interventions motivate participants in engaging in the activity and results in more critical reflection. Earlier examples of these can be found in [7, 10, 11, 15, 20].

3 Approach

This study consisted of a pilot study of 8 months (from September 2023 until April 2024) at the Province of North-Holland. The Dutch provinces are the administrative layer between the national government and the municipalities.¹ The region North-Holland has nearly 3 million citizens² and the organization has around 1,400 employees.

3.1 Pilot study

To gain initial insights into usage, a poll was conducted among employees in April 2023, asking how often they used AI tools such as ChatGPT for their work activities. The poll, which included more than 600 employees, revealed that 1 in 6 were already using ChatGPT or similar tools. This led to discussions about ethics and safety within the organization, although curiosity and applications were also evident. Management and the executive board acknowledged that the development is inevitable and stressed the need to experiment with the technology in a safe manner to properly assess its impact on work. This was the starting point for this pilot study, which tries to investigate that impact from the bottom up. The approach consisted of three components:

Standpoint and guidelines. The guidelines for AI-usage in the organization are detailed in a 'guidance document' for civil servants. These were established by internal IT/AI- policymakers and advisors and largely based on an advisory document from the ethical committee for Dutch Provinces [12]. The guidance document includes a brief description of LLM-technology, well-known tools, applications, legal frameworks, the restriction on the use of certain types of information, the pilot approach, and links to additional information.

Learning and experimentation. Through a pilot study, knowledge and experiences were exchanged while experimenting with genAI-tools. We developed an interactive approach in which employees can learn and share experiences and opinions. Both employees with and without experience and/or knowledge of AI participated. The objective was to better understand the technology's impact on work by evaluating its benefits, drawbacks, opportunities, and risks and thus to foster ethical reflection among civil servants.

Prior to the experiments, four general genAI applications were identified, which were based on literature, existing organization guidelines and known application areas within the organization:

- (1) Improving existing texts – tasks such as translating, making texts more readable, or changing the tone of existing content [21];

¹The Netherlands consists of twelve provinces, each with its own administration and legislative body. Provinces are responsible for spatial planning, regional mobility, economy, road maintenance, and oversight of municipalities, among other things.

²Statistics about North-Holland, such as the amount of citizens, can be found: <https://cijfers.noord-holland.nl/jive>

- (2) Assisting with reading – summarizing or answering questions about (large volumes of) texts [1];
- (3) General information retrieval – requesting information, receiving explanations, brainstorming, and having conversations [9];
- (4) Generating new texts – drafting a first concept of a letter, policy document, or plan [25].

These applications served as a starting point for discussions about genAI. We also used them when designing the serious game and included them in a survey.

Consulting the organization. A limited number of employees within the organization are reached with the ‘learning and experimentation’ pilot. Considering the discussions and potential impact of this technology, a survey was conducted to gather the opinions and reflections who did not participate in the pilot.

3.2 Guided experimentation

First, we developed an interactive introduction to genAI for interested teams. We facilitated pilot participants with a business account on OpenAI to make use of the latest AI models.³ The workshop lasted two hours and focused on knowledge and experience sharing. We discussed the organization’s approach, discussed the opportunities and risks associated with the technology, provided an explanation of AI and LLM technology, and delivered a demo with tips on how to work with it. We then practiced with a specific use case. At the end, participants set their goals for a six-week experimentation period. During these workshops we observed participants interacting with genAI tools, enriching our view of the different awareness levels and modes of interaction with these tools.

3.3 Serious game

To foster ethical reflections and conversations around the use of genAI, we developed a scenario-based card game, called ‘Moralis Machina’. We opted specifically for designing a serious game to make complex topics tangible, promote an exchange of ideas and experiences, stimulate active participation, while teams also learn more about the technology and each other’s perspectives and opinions. The aim of the game is to help governmental teams reach consensus and formulate standpoints on how to (not) use genAI tools. The game consists of concrete potential situations of genAI use in the workspace and describe ethical, organizational and societal challenges. The scenarios are categorized by application, as they were previously identified. For each of these applications of genAI, we created a ‘role’ or metaphor, to make the use of AI more vivid and understandable for non-AI experts. For instance, ‘The Editor’ is the functionality of genAI that can help the user improving existing texts.

With each of these functionalities of genAI, 3 to 5 scenarios were developed (15 in total). The main topics for the scenarios came from the first sessions with teams and the ethical committee’s advisory report of Dutch provinces identifying the risks of genAI [12]. The

³A paid business account is a safer alternative compared to free publicly available tools on the internet, as OpenAI for instance ensures that data enrolled into their platform will not be used for training new models. For the pilot study, this safeguard was enough to use it in the guided experimentation. The organization is currently investigating and building its own applications to offer security and safety in the long-term.



Figure 1: The main and relevant applications of genAI that were distinguished. Metaphors were used to make complex concepts easy to understand and remember for laypersons

scenario cards cover topics such as information reliability, data security, biases, inclusivity, collaboration, and transparency. The cards consist of a scenario and two questions. The first question is more general, while the second is more specific, intending to steer the discussion towards formulating action points under the given scenario. We also included values with each scenario card to make participants aware of the larger (ethical) themes at play in the conversations. In some approaches to discuss ethics, values are the starting point for a discussion. We consciously made the choice to start with scenarios, rather than values, since we reasoned that for a wider audience scenarios will be easier to understand and engage with than abstract values. Including values in the scenario cards, however, helped the researchers to analyze the common patterns in the discussions.

To make the discussions more interactive and steer the dynamics, two more cards were included: the ‘Orator’ and the ‘Dissident’. The ‘Orator’ is the first of the team who speaks up and the ‘Dissident’ is designated to disagree with the team.⁴ Both of these roles can be assigned by the participant picking the card or by the team as a whole. Adding these two cards helps the dynamics of the group discussions in cases when nobody wants to speak up first or when nobody dares to disagree with an existing flow or direction in a conversation.

⁴The dissident is inspired by the ‘designated dissenter’ in [5].

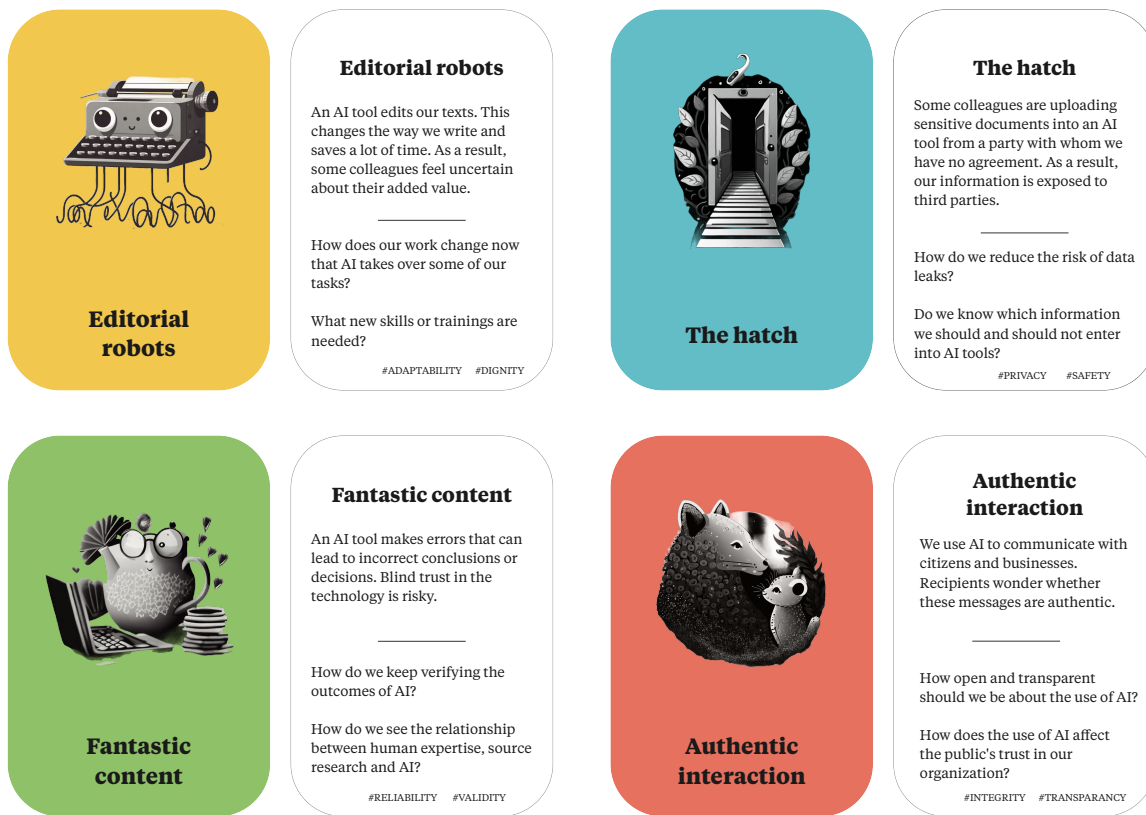


Figure 2: Examples of scenario cards. Each color belongs to a different application of genAI. A scenario card includes a scenario and two questions that enable a team to take position and formulate actions

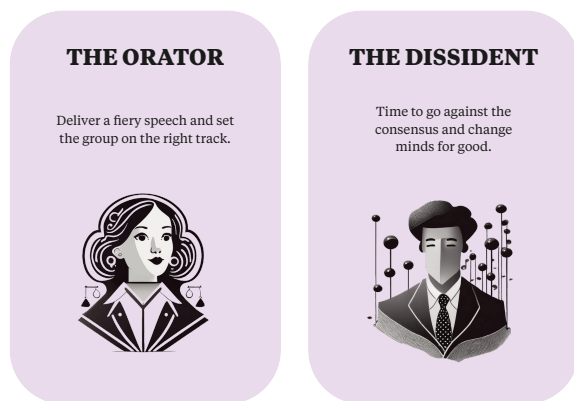


Figure 3: Two additional cards to steer the dynamics of group discussions

3.4 Survey

Finally, to gain better insight into the use of and opinions on genAI tools within the organization, we conducted a survey. The survey

consists of eight general questions (see Appendix A) covering various topics such as AI usage, organizational guidelines, and the impact of genAI on the province's work. The aim is to actively involve employees in the discussion of how the organization should and should not utilize AI.

4 Pilot results

4.1 Guided experimentation

We started the pilot with 5 use cases to assess the impact of genAI on work. However, as we progressed, there was a significant increase in interest. As described, we practiced with specific cases, in which participants experimented with concrete goals. The content of the initial meetings was largely the same, although some parts were adjusted to fit the work and the themes relevant to the participants. In total, 17 introductory workshops were organized, in which over 100 employees took part. The discussions and results were recorded by the researchers on whiteboards, notes and reports. For an overview of the roles and topics, Appendix B can be consulted. In these first workshops we noticed large variations in knowledge and awareness levels between participants. We also noticed the large variety of applications of genAI that employees were interested in experimenting with.

After a team ended the experimentation period, we aimed to organize a session to reflect on the use of genAI. Participants were asked to present the goals they had set beforehand, the steps they took to achieve them, and the key lessons they learned. About half of the teams participated in a concluding session. The other teams were mostly interested in the interactive introduction and did not experiment much with genAI tools afterwards. Inquiries by the researchers revealed that the reasons for not experimenting included: high workload, a perceived lack of immediate benefits, uncertainty about AI and general digital skills, and in some cases, resistance to the technology.

We highlight four example use cases below. Each case includes a brief summary of how genAI tools have been used in the experimentation period, findings on what worked well and what did not, and finally, a remark on the possible impact of generative AI on the work of the team involved. These descriptions have been reviewed and approved by the use case participants.

4.2 Insights in the usage statistics

By using a central platform and providing accounts to employees for each session, we gained insight into usage statistics. With this data, we were able to roughly estimate the climate impact of usage. Our analysis shows that during the pilot period, employees typically used the platform briefly. Between September 2023 and April 2024, one-third of the users (45 out of 123) used the environment for only one day. On average, active use lasted 6.5 weeks, with about 10 interactions per user. However, the number of interactions and number of tokens used varied significantly per user.⁵

This allowed us to make a cautious estimate of energy consumption⁶, although it should be noted that many factors influence energy consumption, such as the model type, servers, and the complexity of requests. From September 2023 to April 2024, approximately 120 users used an average of 55,000 tokens, amounting to a total of 800,000 tokens per month. With an energy consumption of 5 joules per token [18], this translates to approximately 1.11 kWh per month. For the entire organization with 1,400 employees, this means a consumption of 13.9 kWh per month. This is comparable to the usage of one desktop workstation of 800W for about 17 hours.

4.3 Serious game

We played the serious game with 9 teams, ranging from smaller teams of 4 or 5 employees to large teams of over 20 employees. The workshop lasted one to two hours.⁷ The outcomes of each session were transcribed in notes, which were qualitatively analyzed; the outcomes were clustered and categorized per theme. As such, 8 main topics were identified: (1) coherence and authenticity in communication, (2) impact on roles and tasks, (3) data leaks and

data protection, (4) speed of work and future of jobs, (5) reliability and critical thinking, (6) creativity and human-AI collaboration, (7) citation and copyright, and (8) prejudices and stereotypes. For these eight topics, we summarized the overall consensus, the differences between teams and the action points that were formulated. The full summary of the insights of conducting the game can be found in Appendix C.

The relevance and responses to these topics varied. Some teams were overall more critical towards genAI use than others. For the topics 2, 3, 4 and 5 there was a large consensus between teams. Overall, teams agreed that AI should always be used as a supplementary tool, with the human in the lead. The output of AI should always be checked and should serve as a mere inspiration rather than as final output. Teams also shared concerns about data protection, especially for publicly available tools. For topics 1, 6, 7 and 8 there were different viewpoints. There were specifically different opinions as to whether AI tools will expand human creativity or whether the tools make employees lazier in the long-term. Furthermore, there were different opinions whether one should be transparent about the use of AI. Some teams claimed that since it is “just another software” we don’t have to mention all tools used in creating output, while others had a strong opinion that there should be a strict requirement to be transparent.

Overall, more variations in opinions were noticed between teams compared to within teams. Regarding the specific applications of genAI that we identified, various teams within the organization had different viewpoints. Communication specialists, for instance, mentioned they do not want to use AI tools as an ‘Oracle’ or ‘Writer’, since they believe the technology is not reliable enough for these functionalities. Remarkably, legal professionals mentioned there is more risk involved in using AI as an ‘Editor’ and ‘Reading Assistant’, due to the sensitivity of the entered information. Thus, the context of use was important in forming specific standpoints regarding the AI usage in the workspace.

Participants described that playing the game helped them to create awareness of the potential risks of genAI use and forming opinions under which circumstances they should and should not use genAI. In some cases, it also helped participants to better understand the potential impact of genAI on their work. Participants mentioned the importance of having such dialogues amongst colleagues, to create a better understanding and formulate shared viewpoints and actions.

4.4 Questionnaire

Finally, we describe the outcomes of the survey, in which 122 employees that were not part of the pilot shared their experience and thoughts about genAI in the workspace. Surprisingly, as compared with the poll we conducted at the beginning of the study, a large majority of these respondents indicated that they rarely or never used AI tools (78.7%). Nevertheless, most respondents (65.6%) supported the idea that it is important for the organization to embrace the use of AI. Strikingly, almost half of the respondents (47.5%) were unfamiliar with the organizations’ guidelines. For those who were familiar a small majority found the guidelines to be clear (60%), of which 24% indicated them to be too restrictive, whereas too liberal according to 16%. With an open question, respondents

⁵While the average is about 10 interactions, some users had over 100 interactions. There is also a large variation in the use of ‘tokens’, small units of text (similar to syllables) that the AI tool processes. Analysis shows that half of the users required only 20,000 tokens for their ChatGPT interactions during this period. Still, the average is considerably higher at 55,000 tokens, due to a relatively small group using a disproportionately large number of tokens.

⁶This estimation only presents the usage on the facilitated OpenAI platform. It is important to keep in mind that we do not have any insight in usage on free and available tools on the internet.

⁷Ideally, the workshop takes two hours to discuss all scenario cards. However, in practice we sometimes worked with limited availability of teams. In these cases, we picked the scenario cards that were most relevant for the team.

Table 1: Example use cases

Developing a spatial vision (in text and images) by design specialists	Handling of service requests by service desk employees
The design team of the province consists of urban planners and landscape architects who bring together various spatial interests and wishes into a coherent story. This team experimented with ChatGPT and Dall-E2 to help establish a spatial vision for North-Holland in 2050. Experimenting with generating text for specific target audiences, formulating titles, and making summaries of existing texts were all considered useful. However, creating new scenarios for specific regions of North-Holland did not work well, as the tools lacked knowledge and context of Dutch municipalities. Furthermore, while the team enjoyed creating images with Dall-E2, the images were not realistic future scenarios, again due to the lack of specific local knowledge. Overall, the design team did not consider the quality of current genAI tools good enough to have a large impact on their work.	Service desk employees handle questions and complaints from citizens and companies in North-Holland. The team experimented with ChatGPT to help formulating responses in answering the questions and complaints. The team considered ChatGPT useful. Both for formulating a draft of a response or editing existing answers and helping to formulate them in a more understandable way, the tool was considered effective. During the experimentation period, the team made daily use of genAI tools, which significantly reduced the processing time of reports and has a considerable impact on their work. The team is aware of the potential drawback of leaning too much on genAI and claims the service desk employee should at all times do a final check and remains accountable for the responses.
Detecting accidents on roads of North-Holland by traffic engineers	Editing of texts by communication and documentation specialists
The province strives for zero traffic fatalities on provincial roads and uses a registration system for accidents. However, accidents or important details are sometimes missing in this system. It has been investigated whether AI can assist in analyzing online news articles to detect unknown information about the location, time, vehicles involved and emergency services. After scraping articles from the web, GPT3.5 was linked to analyze the texts. Various prompts were created to accurately extract information from the articles. Trial-and-error yielded a combination of prompts that enabled GPT3.5 to get most useful information out of the articles. At the end of the experimentation phase, about 100 messages were analyzed for accuracy, where more than 90% were correct. In some cases, the GPT3.5 hallucinated and made up details that were not in the original text. Although still in an experimentation phase, this case shows some potential use of LLMs for traffic engineers, who need to continuously monitor all accidents on the roads.	The province communicates through various texts and outlets. Communication specialists, as well as documentation specialists, experimented in this use case with ChatGPT to modify existing texts, meant for publication both externally and internally. Although ChatGPT cannot yet create fully high-quality products, it aids by providing inspiration, new perspectives, and revision suggestions, thereby speeding up the editing process. Specifically, both teams investigated whether texts would improve by providing ChatGPT with an existing list of 10 language rules (the organization's guidelines for 'Clear Language'). The output of the experiment was received well; ChatGPT clearly seemed to apply the rules to the given input and the new text seemed to have been improved with regards to these rules. However, communication specialists in particular doubt whether genAI tools help in the process of redacting, since it is also time-consuming to prompt these tools, and the texts still require final revision by the specialists.

were asked for suggestions on how the use of AI within the organization could be improved. 30 respondents shared a general need for more information, either with an internal educational website, or by interactive means such as workshops. Finally, the survey showed three main concerns about AI usage: the reliability of AI information (79.5%), privacy and data security (54.9%), and energy consumption and climate (30.3%). Impact on work (roles and tasks) was hardly seen as a concern by this group (5.7%).

5 Discussion

In this pilot study, we investigated a bottom-up approach with guided experimentation and a serious game to foster ethical reflection among civil servants. Participants in the pilot showed an increased understanding of genAI, its functionalities, and associated risks. After experimenting with genAI tools and engaging in interactive workshops, some participants were excited about the possibilities of genAI, while others were discouraged from using it, once their awareness of the associated risks increased.

While there was growing excitement within the organization, most teams only used the technology for a short period. In only one team a significant impact was experienced, where genAI tools can potentially be used for faster and more comprehensive communication and handling of service requests. In other cases, the impact of AI is limited or difficult to assess for now. While most of the survey respondents (78.7%) indicated they used AI tools rarely or never, 65.6% nonetheless underscored the importance of embracing AI within the organization.

The strength of a bottom-up approach within an organization is receiving input from a wide range of employees, leading to a more comprehensive understanding of ethical concerns related to genAI. While there was mostly consensus on the ethical issues, there were also relevant differences between different teams and individuals. In a follow-up study, it would be beneficial to evaluate the effectiveness of a bottom-up approach by conducting questionnaires at the beginning and end of the pilot. Besides the qualitative reporting throughout the pilot, we cannot know for certain how the

proposed interventions altered the use of genAI by civil servants over a longer time span.

Conducting this pilot highlights the growing need for organizations to guide employees in the responsible use of genAI. The emergence of genAI tools – and the wide accessibility of these tools – raises important questions about whether and how organizations should integrate these technologies into the workspace. Who decides how these tools should be used, and who sets the boundaries? The survey of this study showed that employees are often unfamiliar with organizations' guidelines. Furthermore, ethical guidelines have their limitations: they often only engage employees who are already motivated to act responsibly, while others may remain disengaged or confused about how to apply abstract rules in practice. Ultimately, conflicting advice may arise from different sources, such as national regulations, internal policies, or even personal ethical considerations.

A societal discussion is necessary, in which we include all stakeholders, especially also end users, who make use of (publicly available) genAI tools. In this study, we proposed a serious game that helped civil servants to reflect on the ethical considerations of using genAI tools in the workplace. We learned that employees offer practical insights and identify potential (ethical) issues that may not be evident to upper management or IT/AI-policymakers. Employees from different departments also highlighted unique opportunities and risks specific to their domains, leading to more tailored guidelines and ways of working. These points show that navigating the transition brought about by the emergence of generative AI in the workplace should be seen as an iterative process. It should not only be addressed at the management or policy level. Instead, experimentation and insights gathered from the bottom up should continuously inform the discussion and guide organizations in their future endeavors.

While this study has been focused on genAI usage by civil servants in the public services, the lessons learned can be applied more broadly. The pilot's approach of guided experimentation as well as the scenarios and ethical challenges explored in the serious game are relevant to other sectors as well. This study might thus inspire practitioners in other industries in setting up guided experimentation within their own branch or organization as well as designing interventions to foster ethical reflection on genAI use by end-users.

Acknowledgments

The images of 'Moralis Machina' were made with the generative AI functionality of Adobe Firefly.

6 Appendices

A Survey questions

- (1) How often do you use Artificial Intelligence (AI) tools, such as ChatGPT or Bing Chat, for your work? (multiple choice)
- (2) Which AI applications do you find most interesting or useful? (multiple choice, multiple answers possible)
- (3) What is your opinion on our organization's stance and guidelines regarding the use of AI? (multiple choice)
- (4) What are your biggest concerns regarding the use of AI in our organization? (multiple choice, multiple answers possible)
- (5) How important do you think it is for our organization to embrace working with AI? (multiple choice)
- (6) To what extent do you think AI will change your work in the future? (multiple choice)
- (7) In which areas do you think AI can make a positive contribution to our work? (open question)
- (8) Do you have any suggestions or ideas on how we can improve the handling of AI within our organization? (open question)

B Overview of all pilot participants and themes of the workshops

Table 2: Overview of all pilot participants and themes of the workshops

Professional role in the organization	Topic of the workshop
Designers	Creating a regional spatial vision into a coherent story (in text and images)
Communication specialists	Investigating possibilities to edit and improve various types of texts (standard letters, web content, press releases, job postings)
Managers	An interactive introduction in AI. Tailored more towards organizational changes and strategy.
Service desk employees	Improving the handling of questions and complaints (both internally and externally)
Legal affairs	Analysing objections against provincial decisions
Housing policymakers	Discussing possibilities for the policy work of housing advisors
Programmers	Developing scripts or restructuring data
Traffic engineers	Analysing online news articles to uncover information about previously unknown traffic accidents that have occurred on North Holland roads.
Policy officers in Heritage and culture	Investigating possibilities for Heritage and culture
Management assistants	Experimenting for organizational and support tasks
Documentation/Information specialists	Improving an implementation framework and training materials for colleagues
Procurement specialists	Experimenting with an ‘AI procurement assistant’ to advise on procurement tasks
Individual experimentation (3 sessions)	Focusing on personal everyday support, such as a personal reading and writing aid

C Summary of serious game reports

Table 3: Summary of serious game reports

Theme	Consensus	Remarks	Actions
Coherence and authenticity in communication	Most participants recognized the challenge of communicating clearly and understandably to various target groups. Some colleagues believe AI can assist in writing clearly and can, for instance, automatically adjust the language level of a text. Everyone agreed that when using AI, the authentic attitude as a governmental organization must be maintained.	According to communication specialists it would be helpful to make agreements on the writing style to be used in the form of a style library with templates. However, colleagues from other departments mentioned that it might instead be necessary to relax existing guidelines, since outreach messages are not always well received.	Editors(in-chief) must always check the province’s communication guidelines, before texts are published. Furthermore, the guidelines should be revised, by taking into account AI-mediated content and giving clear guidance. Possibly including a style library with templates.
Impact on roles and tasks	The consensus is that AI can partly take over editorial work, but human editors will still be needed for the final check.	Participants were more likely to think that work from other departments can be automated, but do not see their own work at risk. For instance, editors believed their own work is not at risk, while other people at the organization believed the work of editors to be at risk.	A strategy to make sure tasks remain clear, specifically for editors, would be to provide a clear workflow stating where in the writing process, AI could be beneficial, and in which parts of the process the human editor needs to be involved and check the texts.
Data leaks and data protection	During all sessions, participants were aware of the risk of data leaks and understood the importance of data protection. The guideline that in any situation business-sensitive or personal information should not be entered, was clearly understood. Although participants understood this restriction, concerns remained about the security of the platforms.	There were no relevant differences between groups, the awareness and concerns were present in all teams. We do not know whether employees that did not take part of the pilot share the same awareness of this restriction.	The organization should investigate the safety of the tools being used and make agreements with providers about data security. Simultaneously, the topic of data security should be addressed regularly among colleagues, for instance in integrity and security courses, targeted (online) training and in the onboarding of new colleagues.

Table 3: Summary of serious game reports

Theme	Consensus	Remarks	Actions
Speed of work and future jobs	All teams agree that AI should be used as a complementary tool, rather than taking over the work. Nearly all teams doubt whether AI tools will speed up work, as according to most, the quality is not sufficient to do so yet.	For some specific tasks, such as initial brainstorming and repetitive tasks, AI tools might be used to accelerate the work. Currently, it is highly dependent on the digital skills of the employees whether they can take benefit from these tools or not.	Offer training and facilitate low-threshold experimentation. Facilitate sessions for critical reflection.
Reliability and critical thinking	AI can convincingly provide incorrect information, which can lead to unjustified trust. Participants agree that all AI output must be checked. Blind trust is risky and common sense remains crucial. There is a risk that we may oversimplify complex problems.	One team mentioned the risk of generating information 'out of thin air'. Another team mentioned that the team itself should compensate for the 'laziness' of the tool. Multiple teams mentioned checking for correctness is time consuming and might disqualify the tools altogether.	For the organization it is crucial to explore which tools are the most reliable to adopt as well as investigating how to make existing tools more reliable by providing context-specific information (known as 'in-context learning').
Creativity and human-AI collaboration	Brainstorming with AI raises the question of how much work one wants to outsource to AI and what we will be gain or lose. Most participants believe that AI does not diminish learning moments and that they still need to produce their own ideas. Working with AI is mostly seen as a new skill. To get useful output of AI tools requires practice and creativity.	Some participants see opportunities for collaboration, where AI stimulates human creativity, for example when being stuck in thinking or writing or for (quickly) producing new ideas or scenarios. Other participants were concerned the tools make us more lazy in the longterm.	Employees should stay up-to-date on new developments, without mandatory use of AI tools. Develop and share best practices for using AI tools and stimulate creative ways of working.
Citation and copyright	AI-generated texts can unintentionally contain other people's work without attribution. All teams agree: correct attribution is crucial and texts without attribution should not be published.	Interestingly, communication specialists believe copyright should not be of concern, since AI tools should only be used to edit or modify already existing texts. There are varying opinions as to whether the organization should be transparent about the use of genAI.	Some future AI models might be better at indicating where the information comes from than others. Current tools (such as ChatGPT and DALL-E2) are limited in this respect. Keep investigating which tools are best at citation.
Prejudices and stereotyping	AI can introduce prejudices and stereotypes into texts and images. When using these tools repeatedly, it can potentially lead to certain uncriticized (political) viewpoints and stereotypes being reinforced.	It was noticed that this problem mostly occurs when using genAI to produce new content from scratch. Specific teams recognized this issue more than others. For instance, teams dealing with mostly factual information recognized this problem less.	A solution that is brought up by multiple teams is a diverse panel looking at the output of the organization on a regular basis. Not only for AI-mediated content, but for all content, it is important to reflect on stereotypes, possible biases and to make outlet messages as inclusive as possible.

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